

EXPERIMENT 13: PERFORM PERSPECTIVE TRANSFORMATION ON THE VIDEO

Aim

To perform perspective transformation on a video using Python and OpenCV.

Algorithm

1. Import OpenCV and NumPy libraries.
2. Open the input video using VideoCapture().
3. Read each frame from the video.
4. Define four source and four destination points.
5. Compute the perspective transformation matrix.
6. Apply cv2.warpPerspective() on each frame.
7. Display and save the transformed video.
8. Release the video objects.

Python Code

```
import cv2
import numpy as np

cap = cv2.VideoCapture("input_video.mp4")

if not cap.isOpened():
    print("Error: Cannot open video")
    exit()

width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = cap.get(cv2.CAP_PROP_FPS)

fourcc = cv2.VideoWriter_fourcc(*'XVID')
out = cv2.VideoWriter("perspective_output.avi",
                      fourcc, fps, (width, height))
```

```

while True:
    ret, frame = cap.read()

    if not ret:
        break

    src = np.float32([
        [0, 0],
        [width-1, 0],
        [0, height-1],
        [width-1, height-1]
    ])

    dst = np.float32([
        [50, 50],
        [width-100, 20],
        [80, height-50],
        [width-30, height-30]
    ])

    matrix = cv2.getPerspectiveTransform(src, dst)

    transformed = cv2.warpPerspective(frame,
                                       matrix,
                                       (width, height))

    out.write(transformed)

    cv2.imshow("Original Video", frame)
    cv2.imshow("Perspective Video", transformed)

    if cv2.waitKey(25) & 0xFF == ord('q'):
        break

cap.release()
out.release()
cv2.destroyAllWindows()

```

Input

Input: input_video.mp4

Output

Output: perspective_output.avi

Result

Thus, the perspective transformation was successfully applied to every frame of the video using OpenCV.